

Discussion for Peer Assessment

Here are some considerations (in blue) Professor Eggar thought you might find useful when assessing your peer's work and in revising your own submission.

Conclusions

Your boss asks you to recommend a standard procedure for selecting future credit card applicants.

Q: Based on internal data alone (*no use of the Eggertopia scores*) and your two models' performance on the *test set*, which method would you recommend, binary classification or linear regression, and why?

The linear regression model gives a point estimate of profitability and a range for 90% confidence in the estimate. The 90% confidence range is over \$12,000 - \$6,168 above and below each estimate – and the average profit per customer is only \$1600. This suggests the regression model is not very reliable.

And – the regression model still requires choosing a method to decide whether or not to issue a card. So some kind of binary classification still needs to be made!

One possible approach is to choose all applicants whose point estimate of profitability is positive. This will include many high-risk applicants.

Another approach is to exclude anyone who has a certain percent chance of losing money for the bank. For example, it would be acceptable to exclude anyone who had a 25% chance or higher of generating losses. However, because the error on the linear regression is quite large, this approach will likely exclude more people than the linear regression model.

A third approach would be to pick the 75% of applicants with the highest projected profitability. This method should perform about as well as the binary classification. Exactly how it performs relative to the binary classification could be determined by testing both on the new, validation set (which also includes profitability data).

Hybrid approaches can be considered – although not explicitly addressed in the question. For example, the binary model can be used to select non-defaulters, and then classified defaulters can be reviewed again to see if they fall in a potential high profitability category under the linear regression model, in which case they can be “added back” to the group issued credit cards.

Hint: The original estimates of losses per FN and losses per FP need to be replaced with averages taken from the actual training set profitability data now available – and the optimum threshold re-calculated on the training set.

AN important point to consider is that the approximate profitability per non-defaulting customer, earlier estimated at \$2500 by the bank, turns out to be closer to \$3500.

The average loss for defaulting customers, earlier estimated at -\$5000, turns out to be closer to -\$4000.

The threshold on the binary classification will need to be changed in light of this new, more accurate data, and a new average cost per test calculated.

Test your above recommendation on the (newly provided) Validation Set.

Q: Would your original advice above prove correct for the Validation Set? If not, why not?

This depends largely on how accurate the original binary classification model is. It is acceptable to realize that the original binary model was over-fit to both the training set and the test set. The linear regression model should perform on the validation set similarly to how it performed on the test set.